If we write the equivalent system of equations using the echelon form, we get

$x-y+z=-9, \quad y-z=22, \quad z=-23, \quad 0=-11.$

The last equation is a contradiction.

So the given system of equations is inconsistent and has no solution.

By Rouché - Capelli theorem, we have the following rule:

- If there are n unknowns in the system of equations and $\rho(A)=\rho([A\,|\,B])=n$, then the system $AX=B$, is consistent and has a unique solution.
- If there are n unknowns in the system $AX=B$, and $\rho(A)=\rho([A\,|\,B])=n-k, k\neq 0$ then the system is consistent and has infinitely many solutions and these solutions form a $k-$parameter family. In particular, if there are 3 unknowns in a system of equations and $\rho(A)=\rho([A\,|\,B])=2$, then the system has infinitely many solutions and these solutions form a one parameter family. In the same manner, if there are 3 unknowns in a system of equations and $\rho(A)=\rho([A\,|\,B])=1$, then the system has infinitely many solutions and these solutions form a two parameter family.
- If $\rho(A)\neq\rho([A\,|\,B])$, then the system $AX=B$ is inconsistent and has no solution.

Example 1.33

Find the condition on a, b and c so that the following system of linear equations has one parameter family of solutions: $x+y+z=a,\ x+2y+3z=b,\ 3x+5y+7z=c.$

Solution

Here the number of unknowns is 3.

The matrix form of the system is $AX=B$, where $A=\begin{bmatrix}1&1&1\\1&2&3\\3&5&7\end{bmatrix}, X=\begin{bmatrix}x\\y\\z\end{bmatrix}, B=\begin{bmatrix}a\\b\\c\end{bmatrix}.$

Applying elementary row operations on the augmented matrix $[A\,|\,B]$, we get

$$[A\,|\,B]=\left[\begin{array}{ccc|c}1&1&1&a\\1&2&3&b\\3&5&7&c\end{array}\right]\xrightarrow{\substack{R_2\to R_2-R_1,\\ R_3\to R_3-3R_1}}\left[\begin{array}{ccc|c}1&1&1&a\\0&1&2&b-a\\0&2&4&c-3a\end{array}\right]$$

$$\xrightarrow{R_3\to R_3-2R_2}\left[\begin{array}{ccc|c}1&1&1&a\\0&1&2&b-a\\0&0&0&(c-3a)-2(b-a)\end{array}\right]=\left[\begin{array}{ccc|c}1&1&1&a\\0&1&2&b-a\\0&0&0&(c-2b-a)\end{array}\right].$$

In order that the system should have one parameter family of solutions, we must have $\rho(A)=\rho([A,B])=2$. So, the third row in the echelon form should be a zero row.

So, $c-2b-a=0 \ \Rightarrow\ c=a+2b.$

Example 1.34

Investigate for what values of λ and μ the system of linear equations

$x+2y+z=7,\ x+y+\lambda z=\mu, x+3y-5z=5$

has (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

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